MILITARY SPECIFICATION

BATTERY, DRY BA-1579/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. - This specification covers a zinc-potassium hydroxide mercuric oxide battery used as the power supply for the M56 mine fuse.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-6858	Welding, Resistance: Aluminum magnesium, Non-Hardening Steel or Alloys, Nickel Alloys, Heat-resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-B-55521	Batteries, Dry Packaging and Packing of

STANDARDS

FEDERAL

FED. STD. No. 595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition

FSC 6135

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(Copies of specifications, standards, specification sheets, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Classification of requirements.

Requirement	Paragraph
Selection of specifications and standards	3.2
First Article	3.3
Parts, materials and processes	3.4
Design and construction	3.5
Insulation resistance	3.6
Capacity	3.7
Vibration	3.8
Mechanical shock	3.9
Labeling and marking	3.10
Workmanship	3.11

3.2 Selection of specifications and standards. - Specifications and standards for necessary commodities not specified herein shall be selected in accordance with MIL-STD-143.

3.3 First article. - Batteries furnished under this specification shall be a product which has been subjected to, and passed the first article inspection specified herein. First article samples shall have been manufactured using the same production processes, procedures, and equipment, which are to be employed in fulfilling the contract requirements. All materials and components shall be obtained from the same sources of supply as will be used during regular production.

3.4 Parts, materials, and processes: general. - Parts, materials, and processes shall be such as to enable the product to meet its performance and other requirements, as specified herein. Ratings and derating values of parts and materials shall conform to requirement 18 of MIL-STD-454.

3.4.1 Metals. - All metals which do not enter into the basic electro-chemical reaction of the cell shall resist or be treated to resist corrosion when subjected to any test or storage conditions specified herein.

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3.4.1.1 Dissimilar metals. - When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.5 Design and construction. - Battery, Dry BA-1579/U shall be of the design, construction, physical dimensions shown on Figure 1. It also shall consist of the physical and electrical characteristics specified herein.

3.5.1 Dimensions. - All specified dimensions (See Fig. 1) shall be inclusive of any covering or coating that may be applied, and shall remain within the specified tolerances during storage periods and throughout all tests specified herein.

3.5.2 Weight. - The weight of each battery shall not exceed 12 grams.

3.5.3 Battery voltage. -

3.5.3.1 Nominal voltage. - The nominal voltage of the battery shall be 7.3 volts.

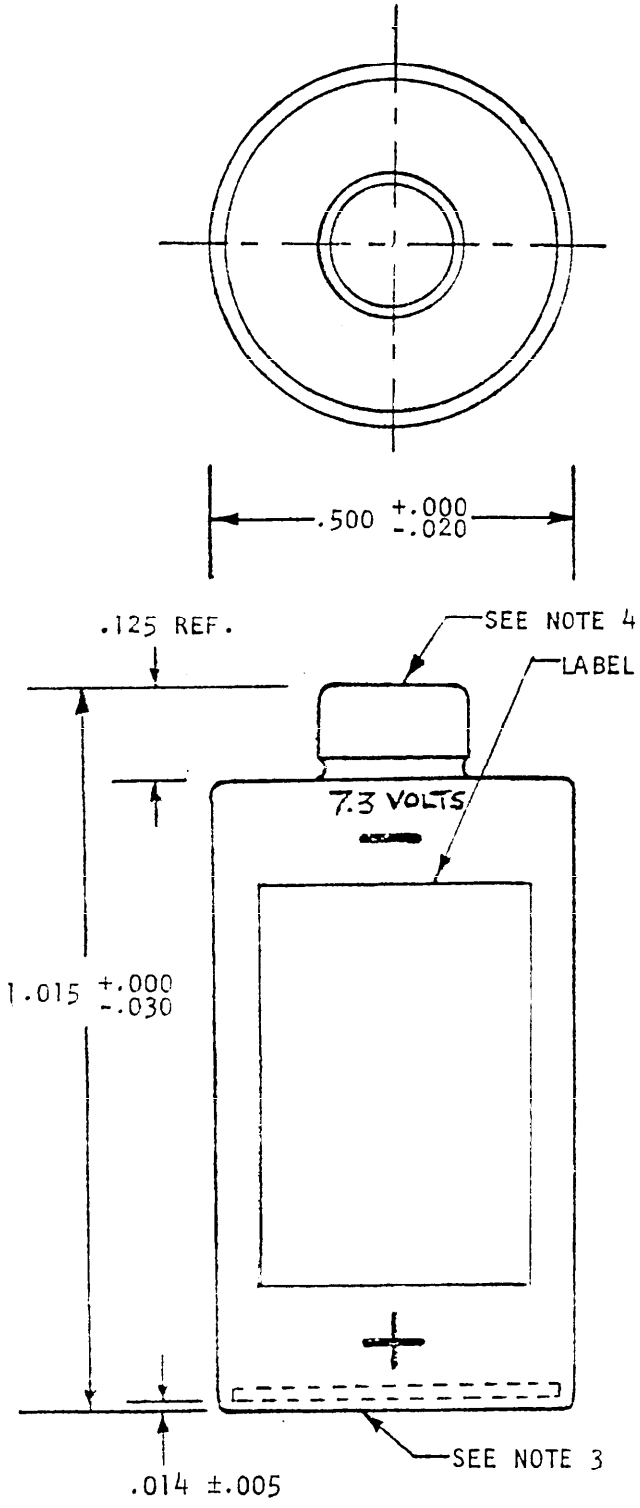
3.5.3.2 Open circuit voltage. - The battery shall have an open-circuit voltage of 7.3 ± 0.5 volts for an initial acceptance requirement at a temperature of 70±10°F (21.1°± 5.6°C). (See 4.7.3.1)

3.5.3.3 Closed-circuit voltage. - The battery shall have a minimum closed-circuit voltage of 6.6 volts after a minimum of five (5) seconds when measured at a temperature of 70° ±10°F. (21.1° ±5.6°C) (See para. 4.7.3.2).

3.5.4 Intercell separation. - A separator shall be placed between cells in this series connected battery. The separator may be an absorbent material. (See para 4.7.4)

3.5.5 Intercell connections. - Welded connections shall be spot welded in accordance with MIL-W-6858, Class B. Connections between cells, terminal and end plate disc shall be so insulated or positioned as to avoid contact with other conducting material in order not to cause a short circuit condition. (See para 4.7.4)

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INCHES	MM
.005	.13
.015	.38
.020	.51
.030	.76
.125	3.18
.500	12.70
1.015	25.78

NOTES:
1. All dimensions shown on figure are in inches.
2. Metric equivalents specified in table above for information only and are based upon 1 inch equals 25.4 millimeters.
3. Nickel disc to be welded to the last cell.
4. Miniature snap-on type terminal per ANSI XVII.

FIGURE 1

BATTERY, DRY BA-1579/U

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3.5.5.1 Disc to cell and intercell weld strength. - The disc to cell weld shall be capable of withstanding the pull of a three (3) pound weight on the disc for a minimum of thirty (30) seconds. (See para 4.7.5)

3.5.5.2 Weld resistance of terminal, disc, and cells. - The weld resistance between the snap-on terminal and the first cell; the weld resistance between the disc and the last cell; or the weld resistance between any two adjacent cells shall be a maximum of 0.1 ohms. (See para 4.7.6)

3.5.6 Age of cells. - The minimum age of cells from the time of their fabrication to the time of their presentation for acceptance inspection as batteries shall be 5 days. The maximum age of cells from the time of their fabrication to their shipping date shall be 90 days. Batteries shall be submitted for acceptance inspection not more than 30 days prior to the shipping date.

3.5.7 Snap-on terminal. - The negative terminal shall consist of a snap-on (miniature type) terminal welded to the top cell as shown in Figure 1. It shall be mounted so that (1) mating member engagements and disengagements will not dislocate the terminal or distort the battery and (2) battery jacket does not interfere with proper terminal mating.

3.5.7.1 Snap-on terminal weld strength. - The welded terminal shall not loosen enough to cause a resistance in excess of 0.1 ohm when tested in accordance with 4.7.7.

3.5.8 Jackets. - The jackets may consist of either metallic or plastic material. The jackets shall contain the cells in such a manner that they fit snugly enough in the jackets to minimize the cell movement. The ends of the cylindrical jacket shall be attached to the cells or crimped to prevent them from slipping out when held or shaken vertically.

3.5.8.1 Color of jackets. - The color of the exposed surfaces of the jacket shall match one of the following lusterless greens 34079, 34086, 34087, 34096, 34102, 34127 and 34128 per Federal Standard No. 595.

3.6 Insulation resistance. - The insulation resistance between each terminal and the jacket of the battery shall not be less than five megohms when tested as specified in 4.7.8.

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3.7 Capacity. - When the battery is tested for capacity for the various specified discharge tests shown in Table I, the time required to terminate the discharge shall not be less than the minimum time specified therein.

TABLE I - Capacity Requirements

Capacity Discharge Test	Method of Test Paragraph	Discharge Temperature	Minimum Discharge Time
Initial	4.7.9	+125° $^{+5°F}_{-0°F}$ ($51.7° \pm 2.8°C$ $^{+0°C}_{-0°C}$)	1/ 18 days
		+32° ± 5°F (0° ± 2.8°C)	1/ 18 days
		-25° $^{+0°F}_{-5°F}$ ($-36.7° \pm 2.8°C$)	1/ 4 days
High Temperature Storage	4.7.10	+113° ± 5°F (45° ± 2.8°C)	2/ 16 days
Long Term Storage	4.7.11	+70° ± 5°F (21.1° ± 2.8°C)	3/ 18 days

1/ The battery shall have a minimum output of 6.00 volts when measured across specified resistance for the first 48 $^{+0.1}_{-0.0}$ hours of operation at all test temperatures. The voltage after the first 48 hours of discharge shall not drop below 6.00 volts at the discharge temperatures and the minimum discharge times specified in the table.

2/ The battery after storage for 90 days $^{+1}_{-0}$ day at the specified temperature and being discharged at the same temperature through the specified resistance shall not fall below 6.00 volts during the minimum discharge time specified in this table.

3/ The battery after storage for 365 $^{+5}_{-0}$ days at the specified temperature and being discharged at the same temperature through the specified resistance shall not fall below 6.00 volts during the minimum discharge time specified in this table.

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3.8 Vibration. - After the batteries have been vibrated as specified in 4.7.12 they shall meet the visual and mechanical, dimensional and closed-circuit voltage requirements. There shall be no open-circuit voltage fluctuations during a 30 second observation period.

3.9 Mechanical shock. - After the batteries have been shocked as specified in 4.7.13 they shall meet the visual and mechanical, dimensional and closed-circuit voltage requirements.

3.10 Labeling and marking. - All labeling and marking shall be clear and legible throughout all the tests specified herein. Labeling and marking shall be black. Battery jackets may have the labeling and marking embossed or die depressed, in which case it may be the same color as the background.

3.10.1 Labels. - Each battery shall have a label positioned as shown in Figure 1. There shall be no information on the label other than the following:

BATTERY, DRY
BA-1579/U
(Contract number) . . .
(Code) . . . . . .
(Lot number) . . . . . .
(Manufacturer's name) . . .
(Plant location) . . . .

3.10.1.1 Code. - The code shown shall indicate the month and year of manufacture of the battery by means of a four digit number in which the first two digits shall indicate the number of the month and the last two digits shall indicate the year. Months earlier than the tenth month shall be represented by a single digit preceded by "0".

EXAMPLES:

A battery manufactured in April 1974 will bear the code "0474". A battery manufactured in October 1975 will bear the code "1075".

When a battery is completed during the last three working days of a month, or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the date to be coded.

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3.10.1.2 Lot numbering. - Lot numbering shall be in accordance with MIL-STD-1168.

3.10.2 Terminal markings. - Terminal markings (both + and -) shall appear on the battery jacket in the positions shown in Figure 1. Markings shall indicate clearly the terminals to which they refer.

3.11 Workmanship. - Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, interchangeability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conforming to prescribed requirements.

4.1.1 Instrument accuracy.

4.1.1.1 Voltmeters. - All voltmeters used in testing batteries shall be of the digital readout type. Vane type voltmeters shall not be used. Digital readout voltmeters shall provide a minimum resolution of 1 millivolt on a 10 volt range with an accuracy to within ±(0.01 percent + 1 digit). The input resistance of digital readout voltmeters shall not be less than 10 megohms on a 10 volt range with a settling time not to exceed 2 seconds to rated accuracy.

4.1.1.2 Ohmmeters. - All ohmmeters used in testing batteries shall be of the digital readout type. Vane type ohmmeters shall not be used. Digital readout ohmmeters shall use the four (4) terminal constant current method and provide a minimum resolution of 10 milliohms with an accuracy to within ±.04 percent (reading + range) on a 100 ohm range.

4.1.1.3 Timing. - Timing equipment shall be accurate within 0.5 percent.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.4)
Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (See 4.5)

c. Quality conformance inspection

(1) Quality conformance inspection of product before preparation for delivery. (See 4.6.1)

(2) Quality conformance inspection of preparation for delivery (See 4.6.2)

4.3 Inspection conditions. - Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80 ± 20°F (26.7° ± 11.1°C), ambient atmospheric pressure and relative

4.4 First article inspection. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.4.1 First article units. - The contractor shall furnish ninety-one (91) first article units of dry battery type BA-1579/U.

4.4.2 First article inspection routine. The first article inspection shall consist of inspections specified in Table II, and shall be performed in the order specified therein. One untested sample battery under Group IV is to remain at the supplier's plant and be available as a reference standard for comparative purposes.

4.4.3 Failure. - If one or more sample batteries fail to meet any of the first article examinations and tests, the sample lot shall be rejected. In the event of rejection, the supplier shall take the necessary remedial action to eliminate the cause of failure.

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TABLE II - First Article Inspection

Group	No. of Batteries	Examination & Test	Req. Para.	Method of Test Para.
I	45	Vibration	3.8	4.7.12
		Mechanical shock	3.9	4.7.13
		Visual-mechanical examination	3.5&3.11	4.7.1
		Open Circuit Voltage	3.5.3.2	4.7.3.1
		Closed Circuit Voltage	3.5.3.3	4.7.3.2
		Initial capacity	3.7	4.7.9
II	30	Visual-mechanical examination	3.5&3.11	4.7.1
		Open circuit voltage	3.5.3.2	4.7.3.1
		Closed circuit voltage	3.5.3.3	4.7.3.2
		High temperature storage capacity	3.7	4.7.10
III	15	Visual-mechanical examination	3.5&3.11	4.7.1
		Dimensions and weight	3.5.1& 3.5.2	4.7.2
		Insulation resistance	3.6	4.7.8
		Open circuit voltage	3.5.3.2	4.7.3.1
		Closed circuit voltage	3.5.3.3	4.7.3.2
IIIA *	10	Snap-on terminal weld strength	3.5.7.1	4.7.7
		Weld resistance of terminal, disc and cells	3.5.5.2	4.7.6
IIIB*	5	Intercell separation and intercell connectors	3.5.4 & 3.5.5	4.7.4
		Disc to cell and intercell weld strength	3.5.5.1	4.7.5
IV	1	Untested-reference sample	-	4.4.2

* Samples for subgroups IIIA and IIIB shall be selected from Group III samples.

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4.5 Inspections covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, materials, and processes - general	3.4
Intercell connections	3.5.5
Lot numbering	3.10.1.2

4.6 Quality conformance inspection.

4.6.1 Inspection of product before preparation for delivery. - The contractor shall perform the inspection specified in 4.5 and 4.6.1.2 through 4.6.1.3.1. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.6.1.1 Lot formation. - The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and is inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which it was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot (1) has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same specifications and drawings, and (2) complies with the provisions for submission of product as specified in MIL-STD-105. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of batteries shall contain cells of one interfix lot number from the manufacturer.

4.6.1.2 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall then be subjected to sampling inspection, utilizing the procedures of MIL-STD-105 with general inspection level II and the AQL's indicated in Table III.

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Table III - Group A Inspection

Examination and Test	Requirement Paragraph	Method of Test Para.	AQL	
			Major	Minor
Visual-mechanical examination	3.5 & 3.11	4.7.1	1.0%	2.5%
Closed circuit voltage	3.5.3.3	4.7.3.2	0.65%	-

4.6.1.3 Group B Inspection. - Group B inspection shall be conducted on each inspection lot of batteries in accordance with Table IV. Tests shall be performed on the same sample batteries in each subgroup in the order shown.

4.6.1.3.1 Group B sampling plans. - The Group B sampling plans for the AQL's listed in Table IV shall be as follows:

Subgroup No.	AQL	Inspection Level
1	0.4%	S-4
	1.5%	S-4
2	0.65%	S-3
3	0.4%	II
	0.65%	II
	2.5%	II

Table IV - Group B Inspection

Examination and Test	Requirement Paragraph	Method of Test Para.	AQL
Subgroup 1			
Dimensions & weight	3.5.1, 3.5.2	4.7.2	0.4%
Snap-on terminal weld strength	3.5.7.1	4.7.7	0.4%
Intercell separation	3.5.4	4.7.4	0.4%
Intercell connections	3.5.5	4.7.4	0.4%
Disc to cell and intercell weld strength	3.5.5.1	4.7.5	1.5%
Weld resistance of terminal, disc and cells	3.5.5.2	4.7.6	1.5%

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Table IV - Group B Inspection (Cont'd)

Examination and Test	Requirement Paragraph	Method of Test Para.	AQL
Subgroup 2			
Insulation resistance	3.6	4.7.8	0.65%
Vibration	3.8	4.7.12	0.65%
Mechanical shock	3.9	4.7.13	0.65%
Open circuit voltage	3.5.3.2	4.7.3.1	0.65%
Closed circuit voltage	3.5.3.3	4.7.3.2	0.65%
Subgroup 3			
Open circuit voltage	3.5.3.2	4.7.3.1	0.4%
Closed circuit "	3.5.3.3	4.7.3.2	0.4%
Initial capacity	3.7	4.7.9	
(a) First 48 hours			0.65%
(b) Remainder of test time			2.5%

4.6.1.4 Group C Inspection. - Group C inspection shall be performed by a Government inspection facility (See 6.1(e)) on inspection lots that have passed Group A and Group B inspection. The inspection shall be conducted in accordance with Table V. Shipment of batteries shall not be withheld pending results of Group C inspection. When the total number of failures is equal to or greater than the rejection number cited in Table V, the Government may refuse to accept batteries for shipment until adequate corrective action has been taken by the contractor as evidenced by retesting.

4.6.1.4.1 Shipment of samples. - Sample batteries selected at random from an inspection lot in accordance with quantities specified in Table V shall be shipped to the Government inspection (See 6.1(e)) at no additional cost to the Government.

4.6.1.4.2 Packaging of samples. - Sample batteries allocated as specified in Table V shall be separately unit packaged as specified in Specification MIL-B-55521 (See 5.1). Each unit package shall be marked

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"RESERVED FOR TEST (*), DO NOT OPEN UNTIL COMPLETION OF STORAGE PERIOD." The asterisk (*) shall be replaced by "1" for subgroup 1 test samples or "2" for subgroup 2 test samples, as applicable.

4.6.1.4.3 Packing of samples. - The unit packages shall be packed in a shipping container in accordance with Specification MIL-B-55521 for Level C Packing (See 5.1).

Table V - Group C Inspection

Test	Requirement Paragraph	Method of Test Para.	Sample Size	Rejection Number
Subgroup 1 High temperature storage (a) First 48 hours (b) 16 days	3.7	4.7.10	200	 4 8
Subgroup 2 Long term storage (a) First 48 hours (b) 18 days	3.7	4.7.11	200	 * *

* Informational test only.

4.6.2 Inspection of preparation for delivery. - Inspection of preparation for delivery shall be accomplished in accordance with the quality assurance provisions of Military Specification MIL-B-55521.

4.7 Method of examinations and tests. -

4.7.1 Visual and mechanical examination. - Batteries shall be examined externally to determine compliance with all applicable requirements and characteristics listed in Table VI.

Table VI - Visual and Mechanical Examination

Requirement	Reference Para.	Type of Defect
Jackets	3.5.8	Major
Color of Jackets	3.5.8.1	Minor
Age of cells	3.5.6	Major
Labeling and marking	3.10	Minor
Labels	3.10.1	Minor
Terminal marking	3.10.2	Major
Workmanship	3.11	Minor

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4.7.2 Dimensions and weight. - Batteries shall be examined by measuring or gaging, and by weighing to determine conformance.

4.7.2.1 Dimensions. - All dimensions shall include any coating which may be used, and shall remain within the specified tolerances through the required tests. If box gages are used, a battery loaded with a five (5) pound weight shall pass freely through its applicable gage openings. The inside dimensions of the box gage shall be the specified maximum outside dimensions of the battery.

4.7.3 Battery Voltage. -

4.7.3.1 Open-circuit voltage. - The battery shall be tested for the open-circuit voltage requirement specified in 3.5.3.2 by using an approved digital readout voltmeter. Testing is non-destructive and accepted batteries shall be returned to the inspection lot.

4.7.3.2 Closed-circuit voltage. - The battery shall be tested for the closed-circuit voltage requirement by measuring the voltage across a resistance of 32,500 ± 100 ohms for a minimum of five (5) seconds. An approved digital readout voltmeter shall be used to measure the voltage. Testing is non-destructive and accepted batteries shall be returned to the inspection lot.

4.7.4 Intercell separation and intercell connections. - Batteries shall be visually examined internally for the specified requirements of 3.5.4 and 3.5.5. Examination is destructive and samples subjected thereto shall not be returned to the inspection lot.

4.7.5 Disc to cell and intercell weld strength. - The battery shall be supported by the negative terminal and held in a vertical position. A three (3) pound weight shall be connected to the disc for a minimum of thirty (30) seconds. There shall be no signs of separation at any welded tab joint. The battery shall be classified defective if it fails to comply with all of the specified requirements of 3.5.5.1. Testing is destructive and samples subjected thereto shall not be returned to the inspection lot.

4.7.6 Weld resistance of terminal, disc and cells. - The jacket shall be removed from the battery and resistance measurements made using an approved digital readout ohmmeter to determine conformance with the specified requirement of 3.5.5. Testing is destructive and samples subjected thereto shall not be returned to the inspection lot.

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4.7.7 Snap-on terminal weld strength. The battery shall be tested for the specified requirement by snapping on and off the negative post using an approved fixture which simulates the mating connector. At the end of the twenty (20) engagements and disengagements the battery shall be removed from the jacket. The resistance between the terminal post and the first cell shall be measured with a digital readout ohmmeter to determine conformance with the specified requirement of 3.5.7.1. Testing is destructive and samples subjected thereto shall not be returned to the inspection lot.

4.7.8 Insulation resistance. - Batteries shall be stored for a period of 48 hours at 70° ± 5°F (21.1° ± 2.8°C) and a relative humidity of 50 ± 15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct-current potential of 500 ± 20 volts between each terminal and the jacket of the battery. The insulation resistance of batteries having a nonmetallic jacket shall be measured by the use of a 1 inch-square copper plate making physical contact with the jacket. The plate shall be placed with the broad surface against any area of any surface of the jacket other than that on which the terminals are located. (See 3.6)

4.7.9 Initial capacity. - The sample lot of batteries to be subjected to this test shall be divided into three (3) equal sublots. One third of the sample batteries is to be tested at $125^{\circ}\ ^{+5^{\circ}F}_{-0^{\circ}F}$ $(51.7^{\circ}\ ^{+2.8^{\circ}C}_{-0^{\circ}C})$, another third to be tested at 32° ±5°F (0° ±2.8°C), and the final third to be tested at $-25^{\circ}\ ^{+0^{\circ}F}_{-5^{\circ}F}$ $(-36.7^{\circ}\ ^{+0^{\circ}C}_{-2.8^{\circ}C})$.

Batteries shall be temperature conditioned for twenty-four plus two minus zero $(24\ ^{+2}_{-0})$ hours at the specified discharge temperature. The battery at the applicable temperature shall be discharged through a resistance of 32,500 ± 100 ohms to a test end voltage of 6.00 volts. Voltage monitoring using an approved digital readout voltmeter shall be performed a minimum of once every minute of run down time, and printout for the first 48 hours shall be at least once an hour. Printout data after the first 48 hours shall be obtained at least a minimum of once every six hours.

4.7.10 High temperature storage capacity. The battery shall be stored for 90 days $^{+1}_{-0}$ day at a temperature of $113^{\circ}\ ^{+2^{\circ}F}_{-8^{\circ}F}$ *$(45^{\circ}\ ^{+1.1^{\circ}C}_{-4.4^{\circ}C})$ and then discharged at this same temperature through a resistance of 32,000 ± 100 ohms to a test end voltage of 6.00 volts. Voltage monitoring, using an approved digital readout voltmeter, shall be performed a minimum of once every minute of run down time, and printout for the first 48 hours shall be at least once an hour. Printout data after the first 48 hours shall be obtained at least a minimum of once every six hours.

* See TABLE VII

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4.7.11 Long term storage capacity. - The battery shall be stored for 365 $^{+5}_{-0}$ days at a temperature of 70°± 5°F* (21.1° ± 2.8°C)* and then discharged at this same temperature through a resistance of 32,500 ± 100 ohms to a test end voltage of 6.00 volts. Voltage monitoring, using an approved digital readout voltmeter, shall be performed a minimum of once every minute of run down time, and printout for the first 48 hours shall be at least once an hour. Printout data after the first 48 hours shall be obtained at least a minimum of once every six hours.

* See Table VII

Table VII - AMBIENT STORAGE CONDITIONS

Type of Test	Normal Conditions		Extreme Conditions	
	Temperature	Relative Humidity (per cent)	Temperature	Relative Humidity (per cent)
High Temp. Storage	113°$^{+2°F}_{-8°F}$ (45°$^{+1.1°C}_{-4.4°C}$)	50±20	88°F (31.1°C) thru 105°F (40.6°C) and 115°F (46.1°C) thru 118°F (47.8°C)	10 thru 30 and 70 thru 90
Long Term Storage	70 ± 5°F (21.1°±2.8°C)	50±20	60°F (15.6°C) thru 65°F (18.3°C) and 75°F (23.9°C) thru 80°F (26.7°C)	10 thru 30 and 70 thru 90

NOTE:

Normal conditions shall be maintained insofar as possible. Deviations from normal conditions are permitted, provided that: (1) the extreme conditions specified above do not exist for more than ten percent (cumulative) of the specified storage period and (2) that at no time are the extreme conditions exceeded.

4.7.12 Vibration. - The battery shall be vertically vibrated with a simple narmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion). The frequency shall be varied at the rate of 1 cycle per second per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95 ± 5 minutes for each mounting position (direction of vibration) of the battery. The batteries shall be vibrated in three

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mutually perpendicular directions, for approximately equal periods. One of the directions of vibration shall be perpendicular to the terminal face of the battery. The battery's open-circuit voltage shall be observed for a minimum of 30 seconds during the last quarter of each vibration period.

4.7.13 Mechanical shock. - The battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be applied in a direction normal to a face of the battery. For each shock the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (G). The peak acceleration shall be between 125 and 175G. The battery shall exhibit its closed-circuit voltage requirement at the end of the shocks.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Preservation, packaging, packing and marking shall be in accordance with Military Specification MIL-B-55521. Preservation and packaging shall be Level A or C with each unit package consisting of twenty (20) individual batteries, and each intermediate container consisting of twenty-five (25) unit packages. Packings shall be level A, B, or C. Levels of packaging and packing shall be specified in the contract or order (See 6.1(c)).

6. NOTES

6.1 Ordering data. - Contractual documents should specify the following:

(a) Title, number and date of this specification
(b) Date for notice of availability for shipment
(c) Applicable levels of preservation, packaging and packing
(d) Designate destination where each battery lot is to be air shipped.
(e) Government inspection facility performing Group C inspection.
(f) Data cards. - Data cards shall be prepared for each lot in accordance with information specified in MIL-STD-1167.

6.1.1 Indirect shipments. - The packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

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6.1.2 Shipment of mixed lots. - When batteries from more than one inspection lot are shipped as a carload, each lot shall be kept separate, and divisions between lots shall be clearly indicated to prevent mixing of lots in transit.

6.2 Awarding of contract. - Contracts will be awarded only to suppliers who guarantee to meet the requirements of this specification. No combining of performance requirements should be undertaken. Bids that offer to guarantee higher capacities will receive no special consideration in awarding a contract. Contracts will be awarded to the lowest bidder on a cost-per-unit battery basis provided that all performance requirements are guaranteed. Failure on a prior contract of a manufacturer's battery to meet any of the performance requirements of this specification will be adequate cause for rejection of bids on this battery until the manufacturer submits certified data providing that---

(a) Action has been taken to eliminate the causes of failures; and
(b) the battery meets the performance requirements of this specification.

6.3 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definite quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

Army-El

Review Activity:

Army-PA

Preparing Activity:

Army - EL

Project No. 6135-A135

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-49041(EL) BATTERY, DRY BA-1579/U

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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